

Josh Talks AI: Transcriber Quality Evaluation

A Data-Driven Quality Control Framework for ASR Datasets

1. Project Overview & Purpose

This project evaluates the quality of human transcribers using objective data signals rather than manual review. The goal is to automatically identify suspicious transcribers who rush tasks, barely review audio, or make minimal effort. This protects the integrity of AI training datasets (like Whisper AI).

2. How the Workflow Operates (Kaise Kaam Karta Hai)

The automated quality control system operates through a sequential data pipeline:

1. Data Ingestion: The system expects the raw transcription_data.csv containing metrics like audio duration, time taken, and edit status.
2. Feature Engineering: It calculates robust behavioral metrics:
 - Time-to-Duration Ratio
 - Edit Rates
 - Median CPS (Character Per Second)
3. Warning Sign Detection: Isolates anomalies, such as extreme low listening times or rapid copy-pasting of AI text without edits.
4. Risk Scoring (Staged Action Plan):
 - GREEN: Normal metrics (Allow work)
 - YELLOW: Single weak signal (Monitor)
 - ORANGE: Multiple warning signals (Restrict & Enhance QA)
 - RED: Persistent severe anomalies (Block/Review)

3. Technology Stack Used (Kya Tech Use Hua)

- Language: Python 3.x
- Data Processing: pandas, numpy (For fast vectorized feature engineering and outlier detection)
- Data Output/Visualization: matplotlib (For plotting metric distributions), openpyxl (For Excel export)
- Architecture: Modular script-based architecture (run_analysis.py) designed for robust pipeline integration.

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4. Current Status & Dataset Availability

This repository contains the complete analytical framework, formulas, and staging logic. Because the actual dataset was not provided, the project employs a 'Data-Unavailable Version' design. It features a fail-safe execution script (`run_analysis.py`) that detects the absence of data, preventing crashes while keeping the methodology perfectly primed for empirical calibration once the real data is supplied. No synthetic or fake data was fabricated, preserving complete professional integrity.